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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
DATA SCIENCE**

SafeShare

AI23421-WEB TECHNOLOGY AND MOBILE APPLICATION

Project Report submitted by

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1.ABSTRACT

SafeShare is a web-based application designed to provide a secure and reliable platform for sharing different types of digital files over the internet. In today's digital environment, file sharing is a common activity, but it often raises concerns about data privacy, unauthorized access, and security risks. The SafeShare system aims to address these issues by implementing secure file transfer mechanisms and controlled access to shared files. The application allows users to upload, store, and share various file formats such as documents, images, videos, and other digital content through a secure interface. It incorporates authentication mechanisms to ensure that only authorized users can access or download shared files. Additionally, the system focuses on protecting sensitive data by applying security measures during file transmission and storage. SafeShare is designed with a user-friendly interface that simplifies the process of uploading and sharing files while maintaining high levels of security. The platform ensures efficient file management, enabling users to share files safely without compromising data integrity or confidentiality. This project demonstrates how secure web technologies can be utilized to create a reliable file-sharing environment for users.

2.INTRODUCTION

With the rapid growth of the internet and digital communication, file sharing has become an essential part of both personal and professional activities. People frequently exchange documents, images, videos, and other digital files through online platforms. However, many traditional file-sharing methods face challenges such as lack of security, unauthorized access, data leakage, and limited control over shared content. These issues highlight the need for a secure and reliable file-sharing system.

The main objective of the system is to ensure that files are shared securely while maintaining data privacy and integrity. The application allows users to upload files, store them securely, and share them with authorized users through controlled access mechanisms. The system focuses on implementing authentication and secure file transfer techniques to prevent unauthorized access and protect sensitive data. By combining security features with efficient file management, SafeShare aims to create a reliable platform for safe digital file sharing. This project demonstrates the use of modern web technologies to develop a secure file-sharing system that can be used in various environments such as educational institutions, organizations, and personal use.

3.OBJECTIVES

3.1 Primary Objectives

- To develop a secure web-based platform for sharing different types of files safely over the internet.
- To ensure data security and privacy by allowing only authorized users to upload, access, and download shared files.
- To provide an efficient file management system where users can easily upload, store, and share files.
- To implement authentication mechanisms to prevent unauthorized access to files.
- To create a user-friendly interface that makes file sharing simple and convenient for users.

3.2 Secondary Objectives

- To support multiple file formats such as documents, images, videos, and other digital files.
- To maintain data integrity during file upload, storage, and download processes.
- To improve accessibility by allowing users to share files from any device with internet access.
- To enhance user experience through simple navigation and easy file management.
- To demonstrate the use of modern web technologies in developing a secure file-sharing application.

4. Project Description

4.1 Overview

SafeSharing is a secure file sharing system designed to allow users to send, receive, and manage files safely within a controlled environment such as a college or organization. The system provides a secure platform where users can register, log in with OTP verification, and share files like images and PDF documents with other authorized users.

The application includes features such as file preview, activity history, notifications, search functionality, dark mode, and sharing files with multiple users. An admin panel is also provided to manage users, monitor file uploads, and remove fake accounts. SafeSharing ensures privacy, organized file management, and efficient communication between users.

4.2 Tasks Handled

The SafeSharing system performs several tasks to manage secure file sharing:

1	User Authentication	Users register and log in with OTP verification.
2	Profile Management	Users can view and update their profile.
3	File Sharing	Users can send files to one or multiple users.
4	File Preview	Shared files like images and PDFs can be previewed.
5	File Receiving	Users can receive and download files from others.
6	Activity History	Shows file name, sender/receiver, date, and status.
7	Search System	Allows users to search files, users, and shared items.

8	Notifications	Users receive alerts when files are shared.
9	Duplicate File Handling	Automatically detects duplicate files and allows rename, replace, or delete.
10	Admin Monitoring	Admin can view users, monitor uploads, and remove fake accounts.

4.3 Methods Used

The SafeSharing system uses the following methods to operate efficiently:

Authentication Method

- OTP verification is used during registration to ensure secure user authentication.

Secure File Sharing Method

- Files are shared only between registered users to ensure privacy and controlled access.

Duplicate Detection Method

- The system checks uploaded files to detect duplicates and provides options to rename, replace, or delete the duplicate files.

4.4 Tools and Technologies Used

The SafeSharing system is developed using the following technologies:

typescript(backend language)	used for developing scalable and maintainable web applications by adding static typing to JavaScript.
React	Used for building the interactive user interface.
JavaScript	Handles application logic and dynamic behavior.

CSS	Used for designing and styling the user interface.
SQLite	Used as the database to store user data and file information.

These technologies work together to create a secure, responsive, and user-friendly application.

4.5 Database Design

The SafeSharing system uses SQLite to store and manage application data. The database is structured using several tables to organize information efficiently.

Main tables in the system include:

- Users Table – Stores user details such as user ID, name, email, password, and verification status.
- Files Table – Stores file information including file ID, file name, file type, upload date, and file owner.
- Shared Files Table – Maintains records of files shared between users, including sender ID, receiver ID, and sharing date.
- Activity Table – Stores user activity details such as file sharing, receiving, and login history.

This database structure helps maintain secure storage and efficient management of files and user activities.

4.6 System Architecture

The SafeSharing system follows a three-layer architecture.

1. Presentation Layer (Frontend)

This layer represents the user interface where users interact with the system. It allows users to perform actions such as login, upload files, share files, and manage their data. The frontend is developed using React, JavaScript, and CSS to create a responsive and user-friendly interface.

2. Application Layer (Backend)

This layer handles the core logic and processing of the application. It manages user requests such as authentication, file uploading, secure file sharing, notifications, and duplicate file detection. The backend is developed using TypeScript, which runs on Node.js to ensure better code reliability, scalability, and maintainability.

3. Data Layer (Database)

This layer is responsible for storing and managing all application data. It maintains information such as user details, file metadata, and activity logs. The system uses SQLite as the database management system to store and retrieve data efficiently.

System Flow

User Interface (React)



Backend Processing (TypeScript with Node.js)



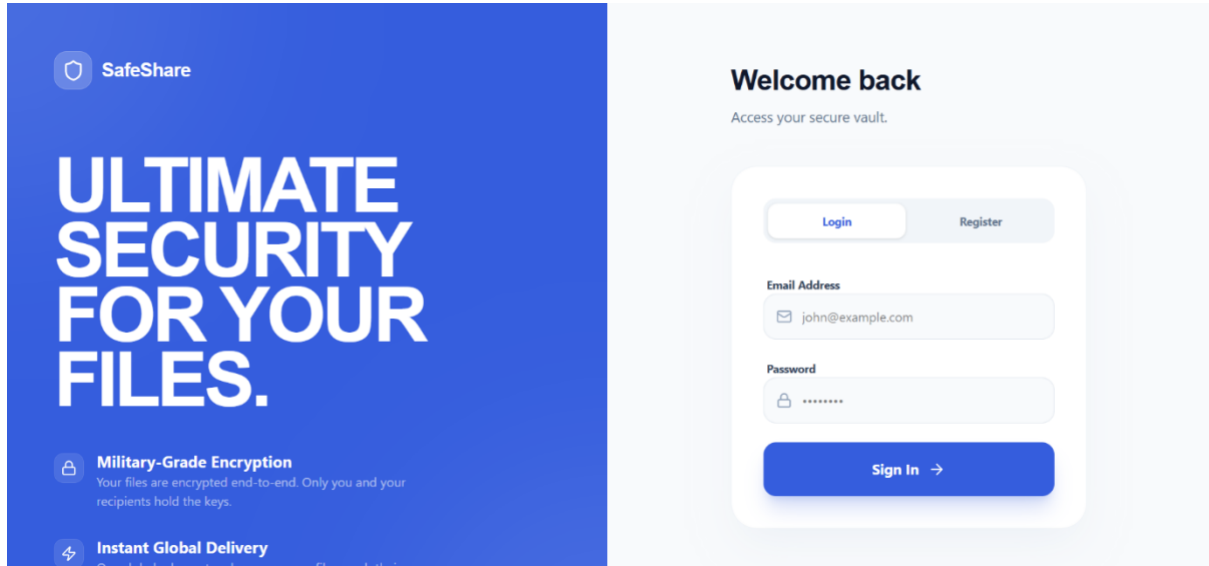
Database Storage (SQLite)

This layered architecture helps in maintaining clear separation between the user interface, business logic, and data storage, making the system more secure, scalable, and easier to maintain.

4.7 Result

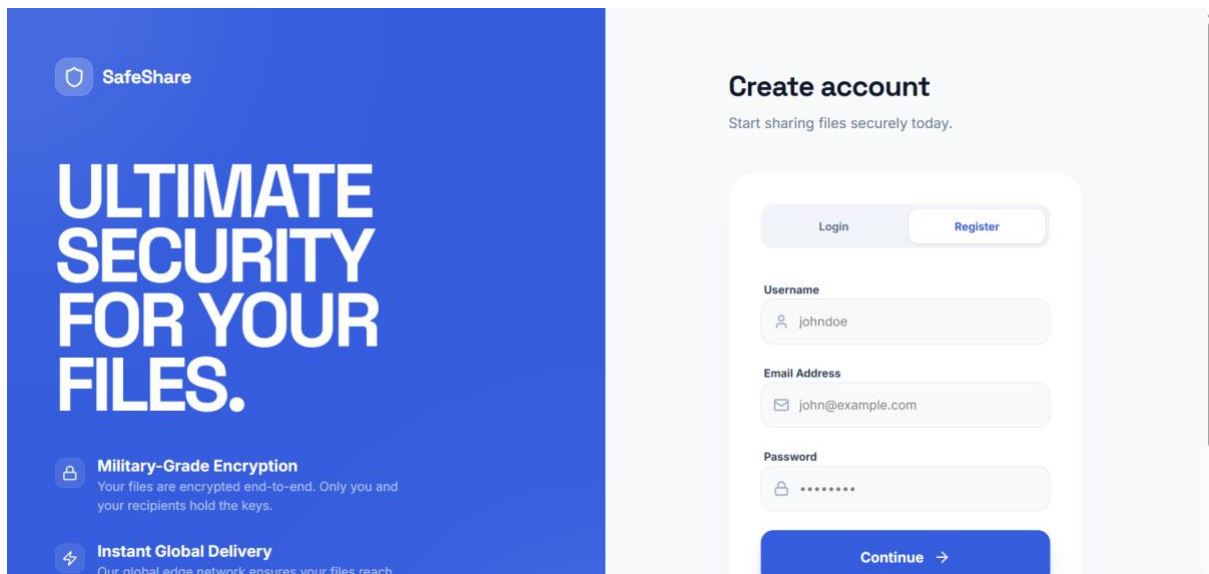
The SafeShare application was successfully developed and deployed with the following outcomes:

Figure 1: Login Page



The login page features a blue sidebar on the left with the SafeShare logo and the text "ULTIMATE SECURITY FOR YOUR FILES." Below this, two features are highlighted: "Military-Grade Encryption" and "Instant Global Delivery." The main content area is light gray and contains a "Welcome back" message, a sub-header "Access your secure vault.", and two buttons: "Login" and "Register". Below these are input fields for "Email Address" (containing "john@example.com") and "Password" (containing "*****"). A blue "Sign In" button with a right arrow is at the bottom.

Figure 2: Register Page



The register page features a blue sidebar on the left with the SafeShare logo and the text "ULTIMATE SECURITY FOR YOUR FILES." Below this, two features are highlighted: "Military-Grade Encryption" and "Instant Global Delivery." The main content area is light gray and contains a "Create account" message, a sub-header "Start sharing files securely today.", and two buttons: "Login" and "Register". Below these are input fields for "Username" (containing "johndoe"), "Email Address" (containing "john@example.com"), and "Password" (containing "*****"). A blue "Continue" button with a right arrow is at the bottom.

Figure 3: Admin Dashboard

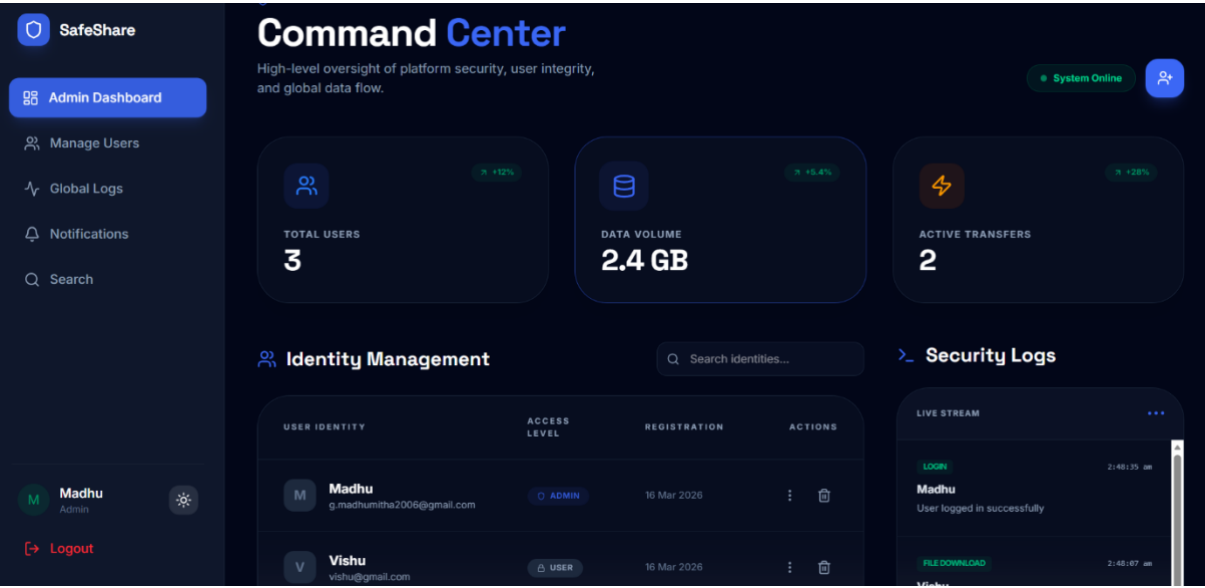


Figure 3.1: Search System

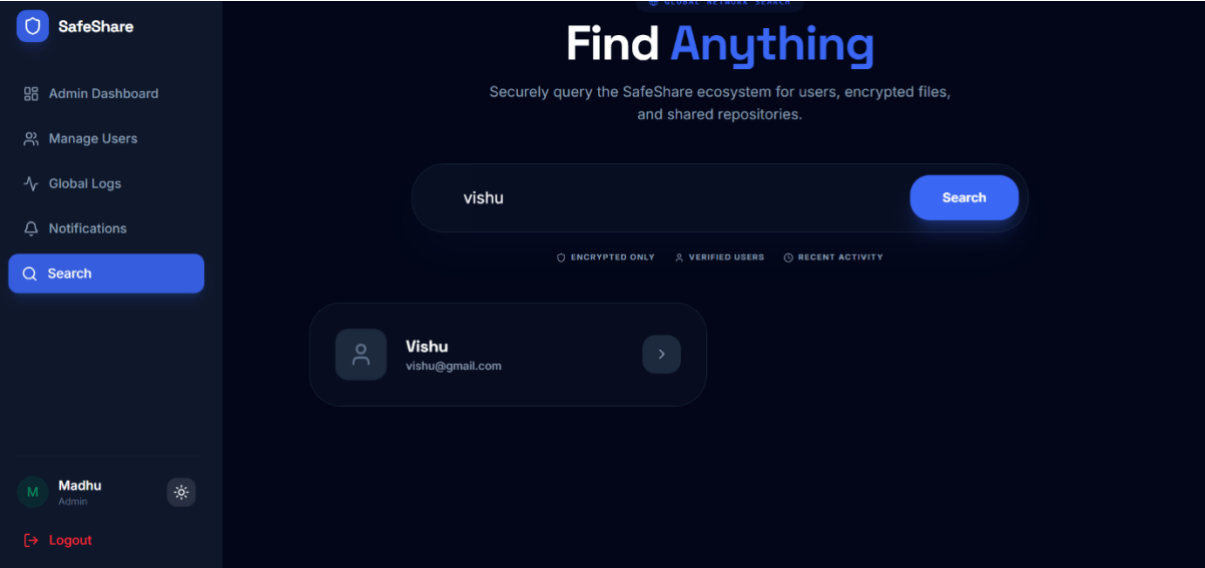


Figure 4: User Dashboard

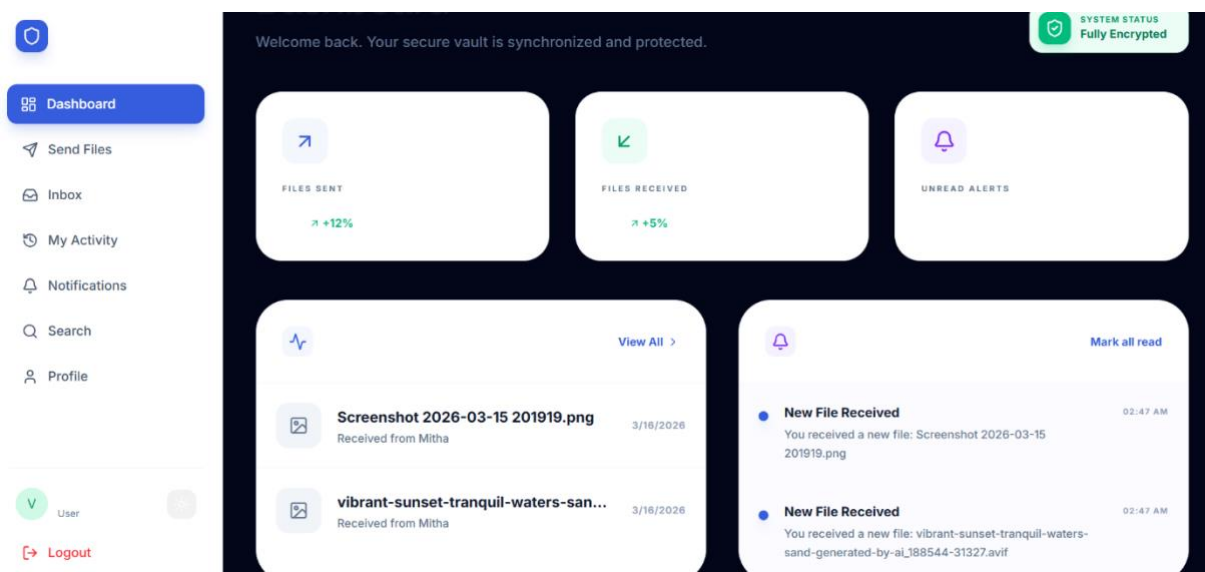


Figure 4.1: Sending Files

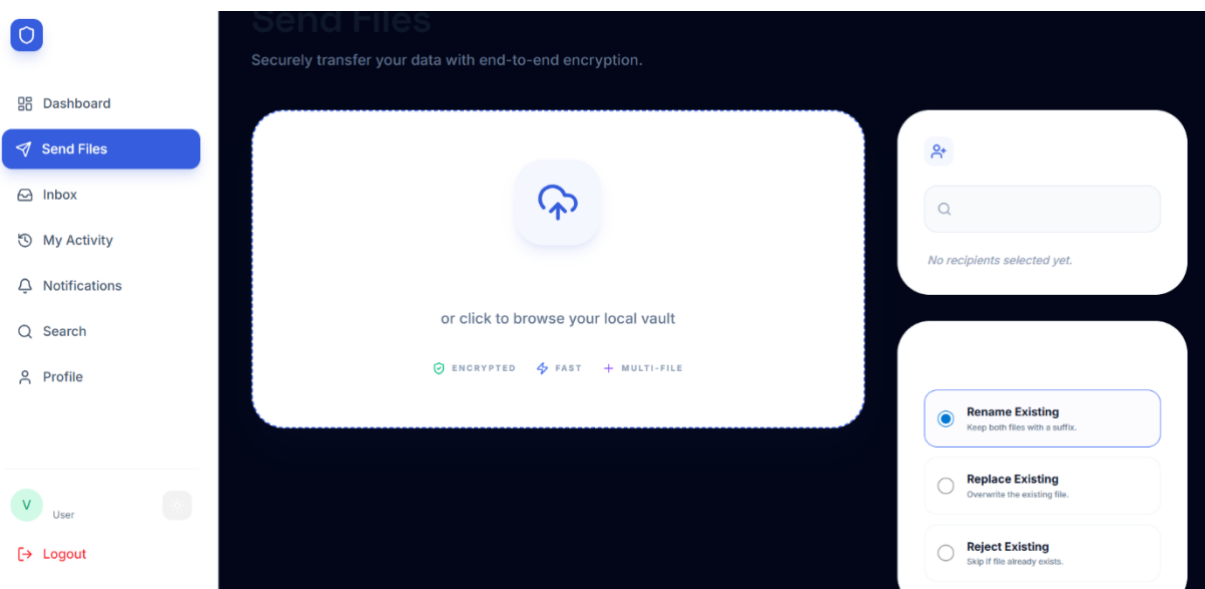


Figure 4.2: Inbox

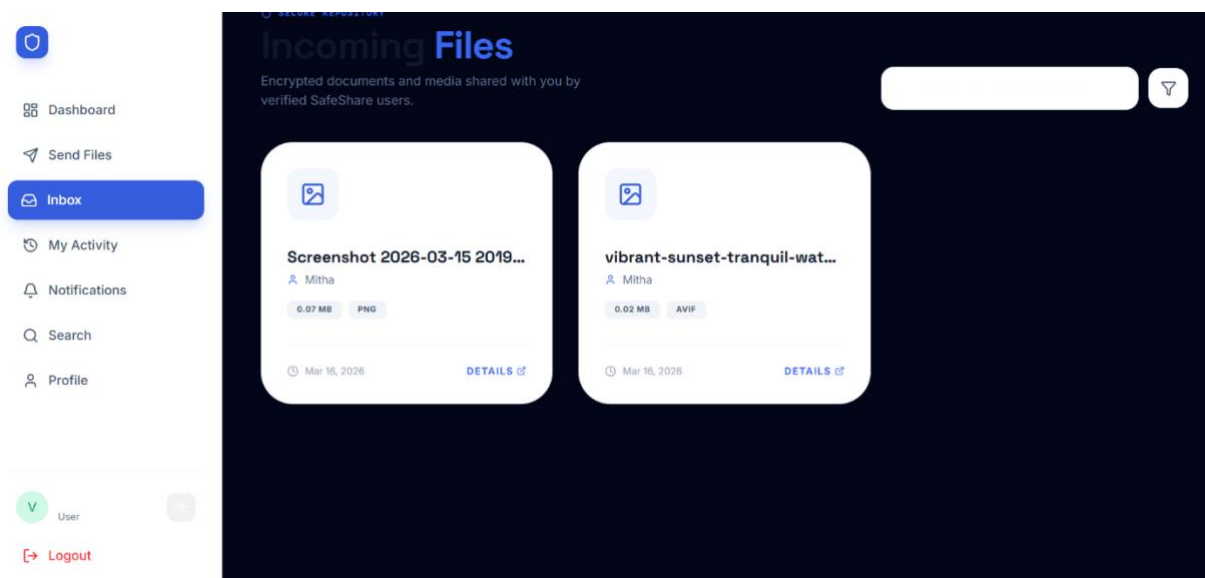


Figure 4.3: My Activity

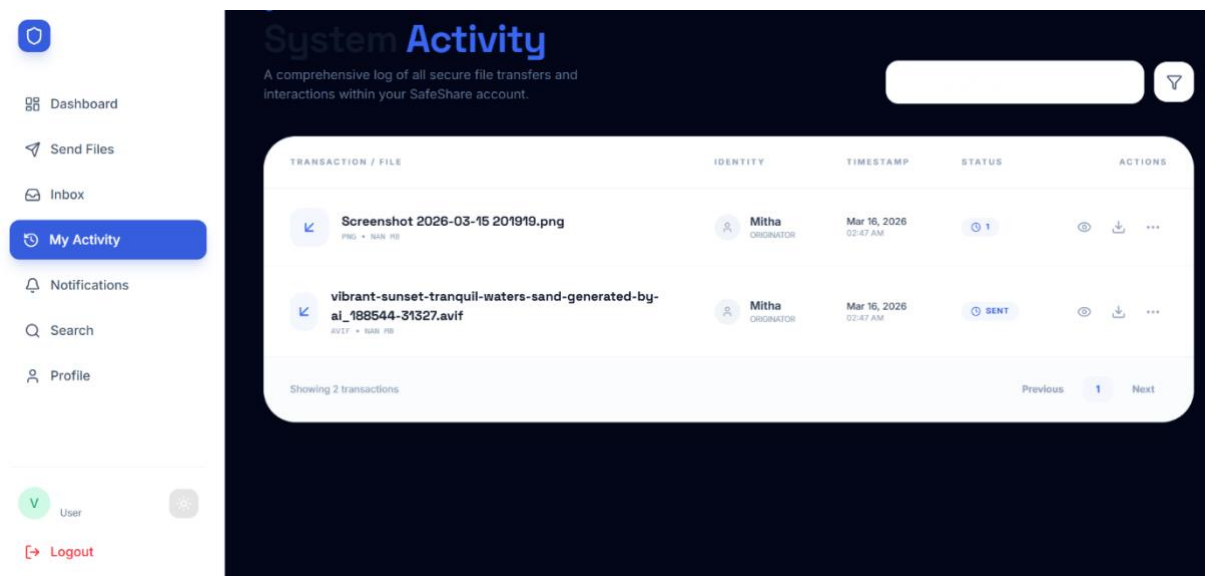


Figure 4.4: Notifications

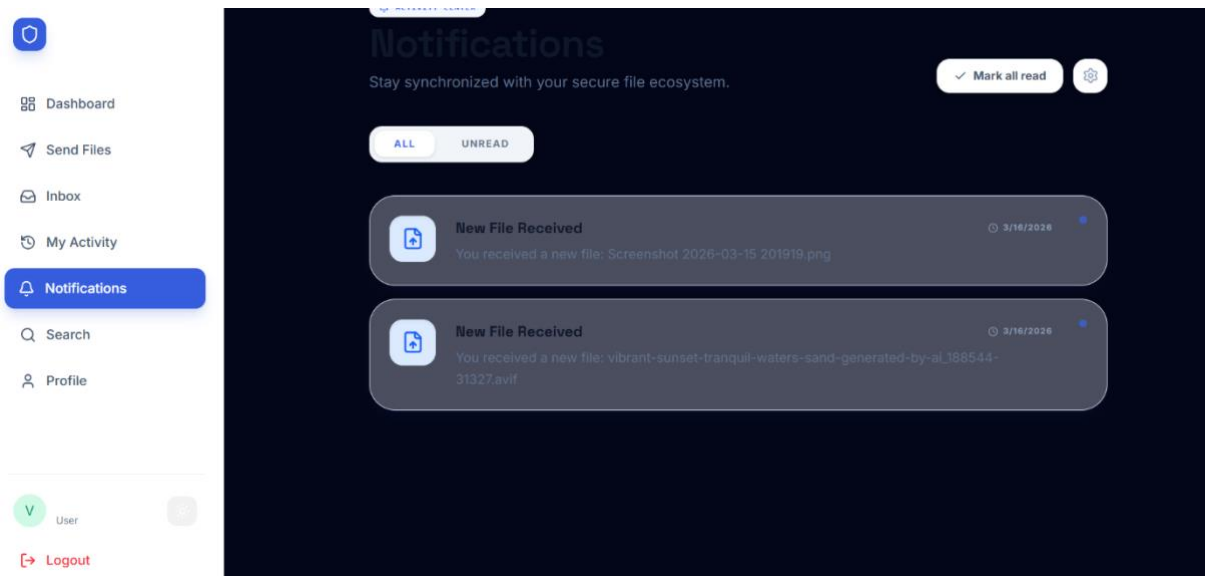
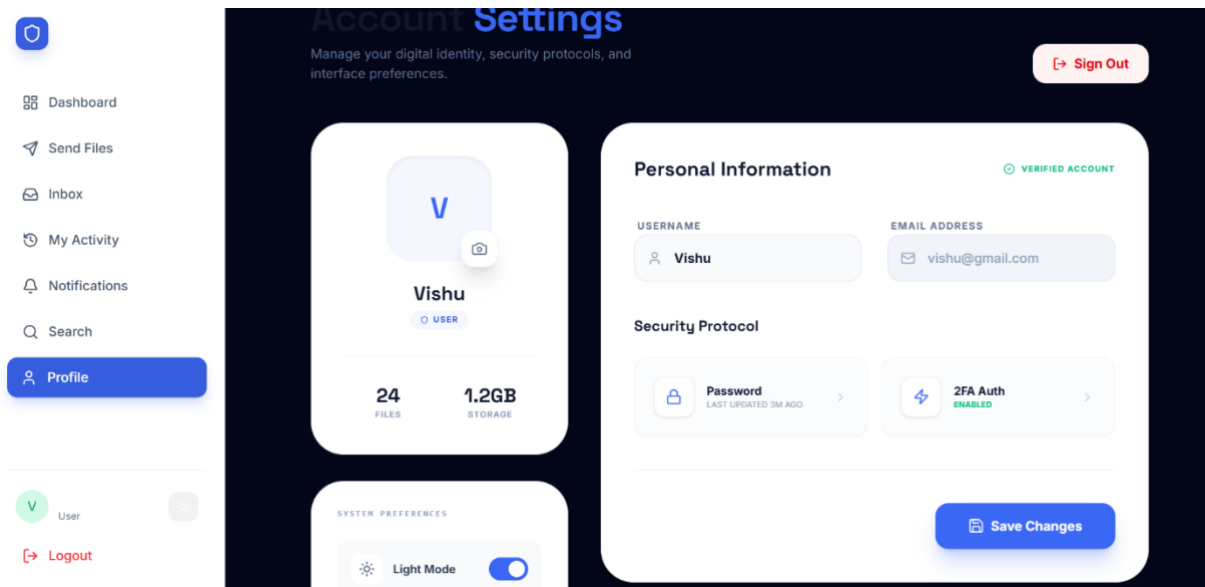


Figure 4.5: Profile



5. Conclusion

The SafeShare web application was developed to provide a secure and efficient platform for sharing digital files over the internet. The system allows users to upload, store, and share different types of files while ensuring data privacy and controlled access. By implementing authentication and secure file management features, the application helps prevent unauthorized access and protects sensitive information.

The system is designed using modern web technologies such as **React** for the frontend, **TypeScript** with **Node.js** for backend processing, and **SQLite** for database management. This combination ensures that the application is scalable, reliable, and easy to maintain.

Overall, SafeShare successfully demonstrates how secure web technologies can be used to build a reliable file-sharing platform. The project provides a practical solution for safe digital file sharing and can be further enhanced in the future by adding advanced security features and cloud storage support.